Saehoon Eo

Dept. of Mathematics, Stanford
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Last update: October 26, 2026

Education

Department of Mathematics, Stanford University Sep. 2024 – Present
Stanford, CA, United States
- Third-year Ph.D. Student
- Advisor: Prof. Jonathan Luk

Department of Mathematical Sciences, KAIST Mar. 2020 – Aug. 2024
Daejeon, Republic of Korea *GPA: 4.28/4.3*
- B.S. in Mathematical Sciences (Summa Cum Laude, Valedictorian)
- Advisor: Prof. Moon-Jin Kang
- Left for mandatory military service (Jul. 2021 – Jan. 2023)

Korea Science Academy of KAIST Mar. 2017 – Feb. 2020
Busan, Republic of Korea *GPA: 4.28/4.3*
- Government-run gifted high school
- Graduated with honors: Valedictorian, Top Honors in Mathematics

Research Interest

- Partial Differential Equations, Kinetic Theory, General Relativity

Awards & Honors

Dean's List, Department of Mathematical Science, KAIST — Fall 2023, Spring 2023, Spring 2021
Honor Student, Qualification for taking high-level graduate courses — Spring 2021 – Spring 2024
Required to take 85 credits within 6 semesters with GPA higher than 3.7
Korea Presidential Science Scholarship, KSAF — Spring 2020 – Spring 2024
Scholarship for honorable undergrad. in science from the Korean government
KAIST Presidential Fellowship (KPF), KAIST — Spring 2020 – Spring 2024
Granted to 30 undergraduates in about 700 freshmen, join KPF honor society
Graduation Award: Top Prize in Mathematics, KSA of KAIST — Feb. 2020
Distinction Award, Singapore International Mathematics Challenge — May 2018

Research Publication

- **Eo, S.** (2026). Nonlinear Phase Mixing in Einstein-Vlasov System near Schwarzschild Spacetime. *Preprint, arXiv:2609.13394*.
- **Eo, S.**, Eun, N., Kang, M. (2026). Global Existence of Classical Solutions to Brenner-Navier-Stokes-Fourier System for Large Data. *Preprint, arXiv:2605.07350*.
- **Eo, S.**, Eun, N. (2025). Traveling wave solutions to a large class of Brenner-Navier-Stokes-Fourier system. *Comm. Math. Sci.*, 24(6), 1577-1593 (2026)
- **Eo, S.**, Eun, N., Kang, M., Oh, H. (2025). Stability of Riemann shocks for isothermal Euler by inviscid limits of global-in-time large Navier-Stokes flows. *Preprint, arXiv:2505.15078*.

- **Eo, S.**, Eun, N., Kang, M. (2026). Stability of a Riemann shock in a physical class: from Brenner-Navier-Stokes-Fourier to Euler. *J. Math. Pures Appl.*, 217, 104007.
- **Eo, S.**, Eun, N., Kang, M., Oh, H. (2025). Traveling wave solutions to Brenner-Navier-Stokes-Fourier system. *J. Differ. Equat.*, 422, 639–658.
- **Eo, S.**, Hwang, S., Yeo, W. (2020). Group Rings Satisfying Nil Clean Property. *Communications of the Korean Mathematical Society*, 35(1), 117–124.

Academic Activity

PDE Perspectives in Gravitational Theory International Centre for Mathematical Sciences	Sep. 07 – Sep. 11, 2026 *Edinburgh, UK*
Thematic Program on Shocks and Singularities The Fields Institute	May. 11 – May. 22, 2026 *Toronto, ON, Canada*
3rd Simons Math Summer Workshop: Partial Differential Equations of Classical Physics Simons Center for Geometry and Physics	Jul. 7 – Jul. 25, 2025 *Stony Brook, NY, United States*
Workshop on Kinetic Theory and Fluids University of Wisconsin-Madison	Mar. 28 – Mar. 30, 2025 *Madison, WI, United States*
New Trends in Hyperbolic Conservation Laws and Related Models Courtyard Marriott Namdaemun	Jun. 23 – Jun. 29, 2024 *Seoul, Korea*
The 10th Korea PDE Winter School Korea Advanced Institute of Science and Technology	Feb. 19 – Feb. 23, 2024 *Daejeon, Korea*
Fluids in Seoul 2024 June E Huh Center for Mathematical Challenges/KIAS	Jan. 15 – Jan. 19, 2024 *Seoul, Korea*
IAS/PCMI Undergraduate Summer School Park City Mathematics Institute — Institute for Advanced Study	Jul. 16 – Aug. 5, 2023 *Park City, UT, United States*
Singapore International Mathematics Challenge NUS High School of Math & Science	May 21 – May 25, 2018 *Singapore, Singapore*

Teaching experience

Teaching Assistant for Math 21: Calculus	Winter 2026
Course Assistant for Math 205A: Real Analysis	Fall 2025
Course Assistant for Math 175: Elementary Functional Analysis	Spring 2025
Course Assistant for Math 115: Functions of a Real Variable	Fall 2024
Tutor for Freshmen: MAS101 Calculus I	Spring 2023

Talks

London PDE Seminar, Imperial College London	Nov. 13, 2026
Analysis and PDE Seminar, UC Berkeley	Oct. 5, 2026
PDE perspectives in gravitational theory, ICMS	Sep. 8, 2026
Partial Differential Equation Seminar, KAIST	Jul. 22, 2026
Analysis, PDE & Probability Seminar, KIAS	Jul. 13, 2026

Other Experience

Department's Student Council — Executive Member Spring 2023 – Spring 2024

- Organize Department Career Seminar for Undergraduates, Fall 2023
- Organize Department Open Lab Event, Fall 2023/Spring 2024
- Mentor for Sophomores, Spring 2023

Republic of Korea Army, 9th div. Jul. 5, 2021 – Jan. 4, 2023

KAIST Undergraduate Math Colloquium — Member of Operating Committee Spring 2021

Summer Camp 2019 Jul. 2019

Jacobs University Bremen — VDI *Bremen, Germany*

- A week of science camp with international participants
- Financially supported by KSA and Ministry of Science and ICT in South Korea